

Supplementary Material: Environment Sensitivity and Reproducibility of Post-BA SfM Benchmarks

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A. What this document supports

This supplementary substantiates four specific statements in the main paper; a reviewer following any of its pointers can go directly to the listed section.

- Protocol, “Track provenance”: “ ~ 0.209 is the reproducible level [for the released RESfM checkpoint] ... unchanged (0.217) under the authors’ exact pinned software stack” (Secs. B–D).
- Protocol: “a stack-replication study (supplementary) shows the no-TTT arms are insensitive to it, while test-time fine-tuned means inherit solver-level nondeterminism” (Sec. E).
- Table 2, † footnote: “TTT is run-to-run nondeterministic” (Sec. E).
- Reading any post-BA mean in this literature: differences $\lesssim 0.05$ are within environment noise (Sec. F).

In addition, Sec. G tabulates the per-scene translation, rotation, and registered-camera (N_r) results behind the main paper’s dataset-level numbers, with dataset-level rotation and registration summaries in Table 11; Sec. H gives the complete training and evaluation configuration, Sec. I provides the per-scene label-degradation breakdown behind the label-ceiling analysis, and Sec. J provides extended versions of the main paper’s Tables 2, 5, 6, and 7 (adding rotation and N_r/N_c) intended to replace them in the camera-ready.

B. The reproduction question

Evaluating the RESfM authors’ released checkpoint in our harness on the 36 MegaDepth test scenes yields a post-BA mean translation error of 0.209 vs. the published 0.169 (rotation 1.89° vs. 1.29°). Since our controlled comparisons hinge on baseline fidelity, we traced this discrepancy to its source. The finding is of independent interest: post-BA means on internet-photo benchmarks carry a practical environment-noise floor of $\mathcal{O}(0.05)$, produced by chaotic basin selection in robust bundle adjustment.

Table 8. Released RESfM checkpoint, MegaDepth 36-scene post-BA means, under three conditions, vs. the published number. N_r is the mean number of registered cameras per scene (mean $N_c=364.6$ ground-truth cameras); the published row averages the paper’s per-scene Table 1 values.

Condition	Trans	Rot (°)	N_r
Published (paper Table 1)	0.169	1.29	298.9
Our env (pip/OpenBLAS), 5-seed mean	0.203	1.82	303.9
Our env, seed 20	0.209	1.89	303.6
Authors’ exact stack (conda/MKL)	0.217	1.92	303.7

C. Eliminating the controllable factors

Data. Our MegaDepth npz files are byte-identical to the authors’ released archive, including the Group-2 300-camera subsamples.

Randomness. Five evaluation seeds vary the pre-BA network output by ± 0.004 and the post-BA mean by ± 0.01 ; negligible.

Network forward pass. The pre-BA output (translation/rotation $\sim 1.73/11.9^\circ$) is essentially seed- and environment-stable; the divergence therefore enters at the robust-BA stage. (Notably, the paper does not report pre-BA numbers; all its tables are post-robust-BA.)

Software stack. We rebuilt the authors’ exact pinned environment from their released environment.yaml: Python 3.8.20, MKL-backed numpy 1.24.3 (mkl=2023.1.0), pycolmap 3.10.0, pyceres 2.3, lightning 2.3.3. (The sole deviation is the CUDA build of PyTorch, forced by GPU-driver compatibility; the BA stage is CPU-only.) Re-evaluating all 36 scenes in this matched environment gives translation/rotation errors of 0.217/1.92°, statistically indistinguishable from our environment’s 0.209/1.89°, and not the published 0.169/1.29°. Registration is equally stable: all three of our conditions register 303.6–303.9 cameras per scene on average (Table 8), slightly more than the published 298.9, so the higher mean error is not an artifact of

065	evaluating fewer cameras.	114
066	D. Mechanism: chaotic basin selection in ro-	115
067	bust BA	
068	Initialization perturbation is at the 10^{-10} level. We	116
069	isolated the BLAS layer by running the identical DLT	117
070	triangulation (the BA initialization) on identical inputs	
071	under OpenBLAS and MKL, single-threaded: over	118
072	11,131 triangulated points the maximum relative dif-	119
073	ference is 4.8×10^{-10} (mean 5×10^{-14} ; NaN patterns	120
074	identical).	121
075	Yet outcomes move by $\mathcal{O}(0.1)$. This sub-nanoscale per-	122
076	turbation changes the per-scene post-BA translation by	123
077	up to 0.167, with 16/36 scenes moving by more than	124
078	0.01. Robust BA on ill-conditioned scenes amplifies	125
079	last-bit differences into different convergence basins.	126
080	The published gap has the same signature. Compar-	127
081	ing our released-checkpoint evaluation to the paper's	128
082	Table 1 per scene: 20/36 scenes agree within 0.05 (8	129
083	essentially exactly), 17 are better in our runs, and the	
084	entire mean gap is carried by four basin flips (scenes	130
085	0046, 0148, 0012, 5016; e.g. 0046: $0.043 \rightarrow 0.892$).	
086	Bidirectional, scene-concentrated flips are the signa-	131
087	ture of basin selection, not of a weaker checkpoint,	132
088	which would shift errors in one direction rather than	133
089	improve 17 of 36 scenes.	134
090	E. Stack replication and TTT run-to-run vari-	135
091	ance	136
092	Beyond the released checkpoint, we re-evaluated both	137
093	from-scratch main-comparison checkpoints under the	138
094	authors' pinned stack (MKL numpy, pycolmap 3.10.0,	139
095	pyceres 2.3), changing nothing else, and additionally	140
096	repeated the TTT evaluation within our own stack.	141
097	The supervised baseline is essentially unchanged across	142
098	stacks: MegaDepth $0.344 \rightarrow 0.350$ mean, $0.070 \rightarrow$	143
099	0.088 median, only 7/36 scenes moving more than 0.05	144
100	(max ± 0.37 , bidirectional). This matches the released-	145
101	checkpoint result above. The SS-RESfM weight+TTT	146
102	arm, by contrast, varies regardless of stack: three eval-	147
103	uations of the same checkpoint (ours, a same-stack re-	148
104	peat, the authors' stack) give MegaDepth means 0.316,	149
105	0.401, 0.453 (~ 18 –20/36 scenes shifting per repeat, up	150
106	to ± 2.9 , largely the same scenes, medians 0.12–0.19).	151
107	The attribution is therefore not the software stack but	152
108	test-time fine-tuning itself: it runs an optimization at	153
109	evaluation time, so the chaotic basin selection docu-	154
110	mented above re-enters the result on every run. Con-	155
111	sequently the paper reports all results from the sin-	156
112	gle declared pipeline, treats TTT means as carrying	157
113	$\sim \pm 0.07$ run-to-run variance, and rests its environ-	158
	and run-independent claims on the deterministic no-	159
	TTT cells (which reproduce exactly on repeat).	160
	F. Implications for reading the paper's num-	161
	bers	162
	(1) The published number and ours are draws from	163
	a distribution over environments; no software match-	
	ing can force a particular draw. (2) Differences	
	of $\lesssim 0.05$ in post-BA translation means on these	
	benchmarks are within environment noise and should	
	not be read as method differences, by any paper.	
	(3) Consequently, meaningful comparisons must be	
	environment-controlled: every number in the main pa-	
	per (all arms, RESfM-from-scratch, and the released	
	checkpoint) is produced in a single environment, and	
	the out-of-distribution margins ($2\times$ and larger) sit far	
	above this noise floor.	
	G. Per-scene results	
	Tables 9 and 10 report the per-scene post-BA trans-	
	lation and rotation errors and registered-camera frac-	
	tions underlying the main paper's dataset-level num-	
	bers (seed 20, the declared pipeline): the ten	
	1DSfM scenes and the 36 MegaDepth test scenes, for	
	the three self-supervised arms (madweight, weight,	
	weight+TTT) and the supervised RESfM baseline	
	trained from scratch under the identical recipe. The	
	per-scene view makes two aggregate statements con-	
	crete: on 1DSfM no single arm dominates, the best	
	mechanism flips scene by scene (with <u>Ellis_Island</u> and	
	<u>Tower_of_London</u> hard for every arm), consistent	
	with the per-scene complementarity analysis in the	
	main paper; on MegaDepth the four arms are close	
	on most scenes and the dataset-level gap is dominated	
	by a handful of high-error scenes. Per scene, the	
	catastrophic scenes (<u>Ellis_Island</u> , <u>Tower_of_London</u>)	
	fail in both metrics, though rotation does not always	
	track translation (weight's <u>Vienna_Cathedral</u> failure	
	is translation-specific: $22.71/0.46$), and N_r stays in	
	the 58–92% range on 1DSfM with no arm systemati-	
	cally dropping scenes. Table 11 completes the dataset-	
	level record with rotation and registered-camera counts	
	alongside translation. Rotation broadly agrees with	
	translation at the dataset level: the supervised Blend-	
	edMVS failure is even larger in rotation (36°) and	
	the high-contamination ordering is preserved, though	
	the best arm can differ per metric (on MegaDepth,	
	weight+TTT leads translation while RESfM leads ro-	
	tation). Registration is contamination-dependent: the	
	supervised baseline registers the most cameras in distri-	
	bution (85% vs. 78–80% on MegaDepth) and on clean	
	Strecha, while under high contamination the label-free	

Table 9. Per-scene 1DSfM results. Post-BA translation and rotation error per scene (camera means; rotation in degrees), registered-camera fraction N_r , and the scene’s ground-truth camera count N_c , seed 20. Best translation per scene in bold.

Scene	madweight		weight		weight+TTT		RESfM		N_c
	Trans/Rot	N_r	Trans/Rot	N_r	Trans/Rot	N_r	Trans/Rot	N_r	
Alamo	1.77/2.45	81%	2.18/2.74	88%	1.67 /2.49	86%	1.73/2.42	87%	577
Ellis_Island	17.59/17.16	79%	15.90 /17.37	75%	17.11/17.74	78%	16.02/18.15	85%	227
Madrid_Metropolis	9.63/5.72	73%	4.37/2.84	88%	3.52/2.35	87%	3.38 /1.82	84%	341
Montreal_Notre_Dame	0.94/0.59	69%	11.35/0.55	72%	1.24/0.81	71%	0.93 /0.59	89%	450
Notre_Dame	0.68/3.33	92%	0.34/1.00	89%	0.33 /0.83	88%	0.49/1.08	89%	553
NYC_Library	2.94/2.32	85%	1.19 /1.16	79%	3.39/10.30	84%	6.28/12.99	86%	332
Piazza_del_Popolo	0.94 /0.43	64%	10.66/19.54	78%	17.05/20.46	75%	2.49/3.15	75%	338
Tower_of_London	76.34/35.61	67%	28.29 /23.30	71%	72.61/17.91	67%	67.95/49.66	63%	472
Vienna_Cathedral	7.30/7.42	64%	22.71/0.46	58%	33.57/26.80	78%	2.29 /1.09	59%	836
Yorkminster	5.79 /8.05	68%	30.87/14.95	85%	10.79/12.20	87%	8.77/6.88	71%	437

arms retain more cameras than supervision (38–43% vs. 37% on 1DSfM-hard) at lower error (madweight).

H. Implementation details

Table 12 lists the full training and evaluation configuration. Both methods share every setting except the base learning rate (10^{-4} self-supervised, 10^{-3} supervised, each selected on its own training objective) and the supervision source (COLMAP-derived labels vs. the adaptive percentile pseudo-labels).

Table 10. Per-scene MegaDepth results. Post-BA translation and rotation error per scene (camera means; rotation in degrees), registered-camera fraction N_r , and N_c , seed 20. Best translation per scene in bold. weight+TTT values are single draws of a run-to-run nondeterministic evaluation (Sec. E).

Scene	madweight		weight		weight+TTT		RESfM		N_c
	Trans/Rot	N_r	Trans/Rot	N_r	Trans/Rot	N_r	Trans/Rot	N_r	
0238	1.41/19.02	84%	0.62/8.23	26%	0.63/8.37	27%	0.12 /1.40	68%	522
0060	0.03/0.29	89%	0.02/0.20	90%	0.37/4.13	86%	0.01 /0.04	94%	528
0197	0.89/7.22	54%	0.38/3.54	39%	1.11/6.94	49%	0.11 /0.97	85%	870
0094	0.45/2.72	65%	0.16 /1.44	66%	0.17/1.34	57%	0.88/5.33	75%	763
0265	3.26/17.62	80%	2.34/5.78	84%	0.19 /0.60	96%	3.09/16.27	74%	571
0083	0.23/4.34	54%	1.56/17.70	80%	0.20/3.20	50%	0.10 /1.40	92%	635
0076	0.38/1.40	76%	0.23/0.61	93%	0.22/0.74	81%	0.02 /0.25	89%	558
0185	0.01/0.21	94%	0.04/0.24	91%	0.09/0.88	87%	0.01 /0.07	96%	368
0048	0.14/1.93	93%	0.02 /0.35	91%	0.03/0.43	88%	0.11/1.85	96%	512
0024	0.18/1.27	84%	0.17 /1.13	82%	1.13/4.78	81%	1.09/3.67	94%	356
0223	0.24/2.66	96%	0.23/1.69	93%	0.30/3.06	93%	0.08 /0.45	95%	214
5016	0.04/0.27	82%	0.02/0.08	89%	0.02/0.07	100%	0.01 /0.08	100%	28
0046	0.22 /2.88	96%	0.30/4.11	96%	0.25/3.88	95%	0.93/14.64	93%	440
0099	0.15/0.81	75%	3.52/2.77	72%	0.13 /1.58	71%	0.39/2.18	84%	299
1001	1.22/1.58	98%	0.73/1.49	98%	0.49 /1.22	98%	0.50/1.23	96%	285
0231	0.15/1.63	93%	0.07/0.64	87%	0.24/1.49	89%	0.02 /0.17	90%	296
0411	0.03/0.22	94%	0.32/2.99	82%	0.46/4.12	84%	0.02 /0.11	92%	299
0377	0.05/0.62	68%	0.26/2.76	75%	0.03 /0.56	82%	0.06/0.82	76%	295
0102	0.02/0.15	97%	0.01 /0.07	97%	0.01/0.08	97%	0.02/0.10	96%	299
0147	2.99/31.69	88%	1.21/15.28	79%	1.20 /15.27	76%	3.22/4.61	83%	298
0148	0.33/2.60	77%	0.11/2.11	70%	0.04/0.62	68%	0.04 /0.94	70%	287
0446	0.03 /0.40	98%	0.19/1.20	96%	0.19/1.04	98%	0.05/0.71	98%	298
0022	0.02/0.14	93%	0.03/0.22	92%	0.04/0.33	90%	0.02 /0.11	93%	297
0327	0.09/1.97	95%	0.02/0.22	96%	0.02/0.21	95%	0.01 /0.09	93%	298
0015	0.30/2.64	69%	0.56/3.87	69%	0.65/3.90	74%	0.30 /1.31	82%	284
0455	0.01 /0.20	95%	0.01/0.21	96%	1.60/0.46	97%	0.09/0.85	98%	298
0496	0.04/0.27	94%	0.04 /0.50	96%	0.09/1.28	97%	0.07/0.46	95%	297
1589	0.06/0.96	89%	0.05/0.28	88%	0.30/1.91	99%	0.01 /0.06	97%	299
0012	0.02 /0.23	94%	0.70/9.51	83%	0.22/2.76	88%	0.34/5.53	96%	299
0104	0.03/0.32	67%	0.18/2.28	68%	0.07/1.44	68%	0.02 /0.30	70%	284
0019	0.17/3.22	78%	0.28/6.55	85%	0.13/3.24	88%	0.01 /0.05	84%	299
0063	0.08/0.63	88%	0.07/0.72	91%	0.06 /0.65	91%	0.07/0.44	91%	293
0130	1.60/10.17	88%	0.16/1.87	65%	0.01 /0.33	65%	0.07/1.03	68%	285
0080	0.05 /1.10	46%	0.20/2.48	52%	0.07/0.70	52%	0.10/0.66	50%	284
0240	0.25 /2.23	92%	0.34/4.64	92%	0.46/5.32	91%	0.35/3.02	95%	298
0007	0.01 /0.07	37%	0.09/1.86	93%	0.14/3.28	90%	0.04/0.99	36%	290

Table 11. Dataset-level translation, rotation, and registered cameras (post-BA, seed 20) for the four main-comparison arms. Per-scene values are camera means; N_r/N_c is the fraction of ground-truth cameras surviving robust BA and the 5-pixel reprojection quality mask, summed over the dataset’s scenes.

	madweight	weight	weight+TTT	RESfM
Translation error (mean / median over scenes)				
MegaDepth	0.422/0.145	0.423/0.182	0.316/0.186	0.344/0.07
1DSfM	12.4/4.37	12.8/11.0	16.1/7.16	11.0/2.93
1DSfM-hard	12.2/8.83	21.9/13.1	26.4/17.7	16.2/13.1
Strecha	2.5/0.97	2/0.026	1.98/0.005	0.006/0.005
BlendedMVS	0.146/0.004	0.1/0.036	0.136/0.011	0.379/0.348
Olsson	2.57/1.15	2.65/1.22	7.74/0.18	8.47/0.17
Rotation error ($^\circ$, mean / median over scenes)				
MegaDepth	3.49/1.33	3.04/1.77	2.51/1.39	2.01/0.84
1DSfM	8.31/4.53	8.39/2.79	11.19/11.25	9.79/2.79
1DSfM-hard	14.40/7.39	25.58/17.70	31.97/12.83	22.70/15.86
Strecha	14.62/8.43	7.48/0.17	7.41/0.02	0.02/0.01
BlendedMVS	9.60/0.38	6.65/3.06	7.91/0.55	36.10/29.55
Olsson	13.57/4.37	13.23/5.41	10.49/2.27	12.34/3.35
Registered cameras N_r/N_c				
MegaDepth	80%	79%	78%	85%
1DSfM	74%	77%	80%	77%
1DSfM-hard	38%	39%	43%	37%
Strecha	91%	96%	98%	100%
BlendedMVS	95%	88%	95%	90%
Olsson	83%	90%	90%	88%

Table 12. Training and evaluation configuration. All arms in the main paper use these settings; the only per-method difference is the base learning rate.

Setting	Value
Training (both methods)	
Epochs	20,000 (one epoch = one pass over all scenes)
Batch	4 scenes
LR schedule	milestones 10k, 11k, . . . , 20k, $\gamma = 0.9$
Base LR	10^{-4} (SS-RESfM) / 10^{-3} (RESfM)
Loss weights	$\alpha_{\text{reproj}} = 1$, $\beta_{\text{cls}} = 1$
Infinity-point margin	10^{-4}
Checkpoint selection	best validation error, evaluated every 500 epochs
Seed	20 (21–22 / 21–24 for the seed studies)
Adaptive pseudo-labels (SS-RESfM only)	
Inlier / outlier percentile	20 / 80 (of per-observation reproj. error)
Threshold separation	scale-aware floor, $0.25\times$ median error
Warm-up	10 epochs before pseudo-labeling starts
Evaluation	
Removal threshold τ	0.6 (0.8 for low-contamination single-camera sets, following RESfM)
Test-time fine-tuning	1,001 iterations, LR 5×10^{-3} , evaluated every 250 iterations
Bundle adjustment	pyceres/pycolmap robust BA, as RESfM

I. Per-scene label degradation

Table 13 gives the per-scene data behind the main paper’s label-ceiling analysis: for each of the 18 scenes we measure the contamination level, the fraction of track observations that receive a COLMAP-derived label at all (coverage), and the precision and recall of those labels against ground truth. The aggregate trend in the main paper is visible scene by scene: below $\sim 10\%$ contamination coverage is 0.92–1.00 and recall is 0.66–0.97 (mostly above 0.8), while above $\sim 15\%$ contamination recall never exceeds 0.50 on any scene (0.25–0.50) and coverage falls as low as 0.37: the supervision signal degrades exactly where outlier handling matters most.

Table 13. Per-scene label-degradation results. For each scene: contamination (fraction of track observations that are GT outliers), COLMAP-label coverage (fraction of observations receiving a label), and precision/recall of the COLMAP-derived outlier labels against GT, sorted by contamination. Recall collapses above $\sim 15\%$ contamination.

Dataset	Scene	Contam. (%)	Coverage	Prec.	Recall
BlendedMVS	58c4bb4f4a...	0.5	1.00	0.86	0.94
Strecha	Herz-Jesu-P8	0.7	1.00	0.51	0.66
Strecha	fountain-P11	0.7	1.00	0.69	0.76
Strecha	Herz-Jesu-P25	2.6	0.99	0.72	0.82
Strecha	entry-P10	2.7	0.99	0.88	0.88
BlendedMVS	5be883a4f9...	4.0	0.98	0.90	0.84
BlendedMVS	5b6e716d67...	6.1	0.92	0.89	0.92
BlendedMVS	5bfd0f32ec...	7.1	0.97	0.97	0.97
BlendedMVS	5b950c7160...	9.2	0.96	0.92	0.94
MegaDepth	0007	11.2	0.73	0.84	0.47
1DSfM	Alamo	26.6	0.76	0.81	0.25
1DSfM	Madrid_Metropolis	28.8	0.67	0.85	0.28
MegaDepth	0041	31.4	0.49	0.80	0.42
MegaDepth	0060	39.0	0.42	0.53	0.45
1DSfM	Piazza_del_Popolo	41.9	0.76	0.90	0.30
MegaDepth	0047	54.9	0.37	0.83	0.50
1DSfM	Ellis_Island	57.9	0.79	0.89	0.37
1DSfM-hard	Gendarmenmarkt	59.8	0.50	0.89	0.29

Table 14. Extended main comparison (extends main-paper Table 2; RESfM-aligned schedule, seed 20). Translation panel repeats the main table (mean / median; $^\dagger/^\ddagger$ as there); rotation (mean / median, $^\circ$) and N_r/N_c panels at seed 20. Best per column in bold.

Arm	MegaDepth (25.4) $N_c=13126$	1DSfM (43.3) 4563	1DSfM-hard (60) 1497	Strecha (1.7) 54	BMVS (3.1) 225	Olsson (0.5) 3858
Translation error (mean / median), the primary metric						
madweight	0.422/0.145	12.39/4.37	17.0 $\pm$ 4.4/ 11.7‡	2.50/0.97	0.146/ 0.004	2.8 $\pm$ 0.3 /1.2 ‡
weight	0.423/0.182	12.79/11.01	21.9 $\pm$ 3.4/12.7 ‡	2.00/0.026	0.100 /0.036	2.8 $\pm$ 0.4/1.1 ‡
weight+TTT	0.39 $\pm$ 0.07/0.14 †	16.13/7.16	21.9 $\pm$ 3.9/14.5 ‡	1.98/ 0.005	0.136/0.011	8.0 $\pm$ 0.4/0.4 ‡
RESfM-scratch (supervised)	0.344/0.070	11.03/2.93	16.22 /13.08	0.006/0.005	0.379/0.348	8.47/ 0.17
RESfM (released)	0.209	10.71	—	0.197	0.330	—
RESfM (paper)	0.169	—	—	—	—	—
Rotation error ($^\circ$, mean / median)						
madweight	3.49/1.33	8.31 /4.53	14.40/7.39	14.62/8.43	9.60/0.38	13.57/4.37
weight	3.04/1.77	8.39/ 2.79	25.58/17.70	7.48/0.17	6.65 /3.06	13.23/5.41
weight+TTT	2.51/1.39	11.19/11.25	31.97/12.83	7.41/0.02	7.91/ 0.55	10.49/2.27
RESfM-scratch (supervised)	2.01/0.84	9.79/ 2.79	22.70/15.86	0.02/0.01	36.10/29.55	12.34/3.35
Registered cameras N_r/N_c						
madweight	80%	74%	38%	91%	95%	83%
weight	79%	77%	39%	96%	88%	90%
weight+TTT	78%	80%	43%	98%	95%	90%
RESfM-scratch (supervised)	85%	77%	37%	100%	90%	88%

J. Extended main-paper tables (camera-ready)

Tables 14–17 extend the main paper’s Tables 2, 5, 6, and 7 with rotation error and registered-camera counts alongside translation, following RESfM’s metric emphasis (translation first, then rotation, then N_r ; N_c last). All values come from the same seed-20 evaluations behind the main paper (rotation panels and N_r at seed 20; translation panels repeat the main-paper cells including their seed bands). In the camera-ready we will replace main-paper Tables 2, 5, 6, and 7 with these extended versions. Reading the additions: rotation broadly agrees with translation (the supervised BlendedMVS failure is even larger in rotation, 36 $^\circ$; the high-contamination ordering is preserved), though the best arm can differ per metric (on MegaDepth, weight+TTT leads translation while the supervised baseline leads rotation). Registration is contamination-dependent: the supervised baseline registers the most cameras in distribution (85%) and on clean Strecha (100%), while under high contamination the label-free arms retain more (38–43% vs. 37% on 1DSfM-hard). In the ablation (Table 16) rotation and registration are statistic-insensitive (7.6–9.0 $^\circ$, 71–74%): the statistic matters through the translation medians. The selection oracle (Table 17) improves rotation as well as translation (MegaDepth 1.22 $^\circ$ vs. the supervised 2.01 $^\circ$) but registers fewer 1DSfM cameras (66%): selecting by translation error can sacrifice registration.

Table 15. Extended full shallow mechanism matrix (extends main-paper Table 5; seed 20; † mean $\pm$ std over seeds 20–22). Translation panel repeats the main table; rotation and N_r/N_c panels: seed-20 means / fractions. Olsson column: means. Best per column in bold.

Arm	MD	1DSfM	Strecha	BMVS	Olsson
Translation error (mean / median), the primary metric					
remove	0.60/0.19	9.1 $\pm$ 2.8/ 1.6 $\pm$ 0.2 †	3.02/1.68	0.27/0.19	2.81
remove+TTT	0.49/0.13	8.7 $\pm$ 1.1 /2.2 $\pm$ 1.1 †	2.97/1.56	0.19/0.07	4.07
hybrid	0.37/0.20	16.0 $\pm$ 1.4/6.7 $\pm$ 4.4 †	2.70/1.39	0.10/0.03	2.82
hybrid+TTT	0.39/0.14	15.4 $\pm$ 1.1/10.7 $\pm$ 3.5 †	3.52/2.65	0.13/ 0.00	3.19
madweight	0.42/0.15	11.5 $\pm$ 2.3/6.0 $\pm$ 1.4 †	2.50/0.97	0.15/0.004	2.57
weight	0.42/0.18	15.8 $\pm$ 3.1/10.9 $\pm$ 2.1 †	2.00/0.03	0.10 /0.04	2.65
weight+TTT	0.39/0.14	17.0 $\pm$ 1.3/10.9 $\pm$ 4.1 †	1.98/ 0.005	0.14/0.01	7.74
RESfm (sup.)	0.34/0.07	11.03/2.93	0.006/0.005	0.38/0.35	8.47
Rotation error ($^{\circ}$, means)					
remove	4.87	5.50	12.96	19.83	12.97
remove+TTT	4.05	5.31	12.81	12.52	15.60
hybrid	2.81	6.98	13.04	7.45	12.54
hybrid+TTT	2.99	11.32	16.62	7.80	13.76
madweight	3.49	8.31	14.62	9.60	13.57
weight	3.04	8.39	7.48	6.65	13.23
weight+TTT	2.51	11.19	7.41	7.91	10.49
RESfm (sup.)	2.01	9.79	0.02	36.10	12.34
Registered cameras N_r/N_c					
remove	71%	59%	96%	87%	80%
remove+TTT	68%	59%	94%	90%	80%
hybrid	78%	82%	67%	90%	88%
hybrid+TTT	77%	80%	76%	95%	89%
madweight	80%	74%	91%	95%	83%
weight	79%	77%	96%	88%	90%
weight+TTT	78%	80%	98%	95%	90%
RESfm (sup.)	85%	77%	100%	90%	88%

Table 16. Extended removal-statistic ablation (extends main-paper Table 6; head reweighting fixed). Translation first, then rotation, then N_r (1DSfM, seed 20).

Statistic	1DSfM trans	med. s20/21/22	1DSfM rot †	N_r	BMVS
MAD (madweight)	11.5 $\pm$ 2.3	4.4 / 7.0 / 6.6	8.0 $\pm$ 0.8	74%	0.15
Huber	11.5 $\pm$ 1.6	3.8 / 12.4 / 5.3	7.6 $\pm$ 0.4	71%	0.11
STD	15.2 $\pm$ 1.2	13.3 / 9.0 / 12.9	9.0 $\pm$ 0.2	74%	0.15

Table 17. Extended per-scene selection (extends main-paper Table 7; mean translation, then rotation, then N_r ; seed 20). The oracle (italics) selects by ground-truth translation error over five mechanisms; its rotation and N_r are those of the selected arms.

Dataset	weight	madwt	contam	oracle (GT)	N_c
Translation error (means)					
MegaDepth	0.423	0.422	0.422	<i>0.116</i>	
1DSfM	12.79	12.39	12.39	<i>5.78</i>	
Strecha	2.00	2.50	2.00	<i>1.98</i>	
BlendedMVS	0.100	0.146	0.100	<i>0.083</i>	
Rotation error ($^\circ$, means)					
MegaDepth	3.04	3.49	3.49	<i>1.22</i>	
1DSfM	8.39	8.31	8.31	<i>5.50</i>	
Strecha	7.48	14.62	7.48	<i>7.41</i>	
BlendedMVS	6.65	9.60	6.65	<i>5.28</i>	
Registered cameras N_r/N_c					
MegaDepth	79%	80%	80%	<i>75%</i>	13126
1DSfM	77%	74%	74%	<i>66%</i>	4563
Strecha	96%	91%	96%	<i>98%</i>	54
BlendedMVS	88%	95%	88%	<i>90%</i>	225